CS 663: Fundamentals of Digital Image Processing
Assignment 3

(Segment, Diffuse, Synthesis, PCA)

Date: 21st October 2026

By

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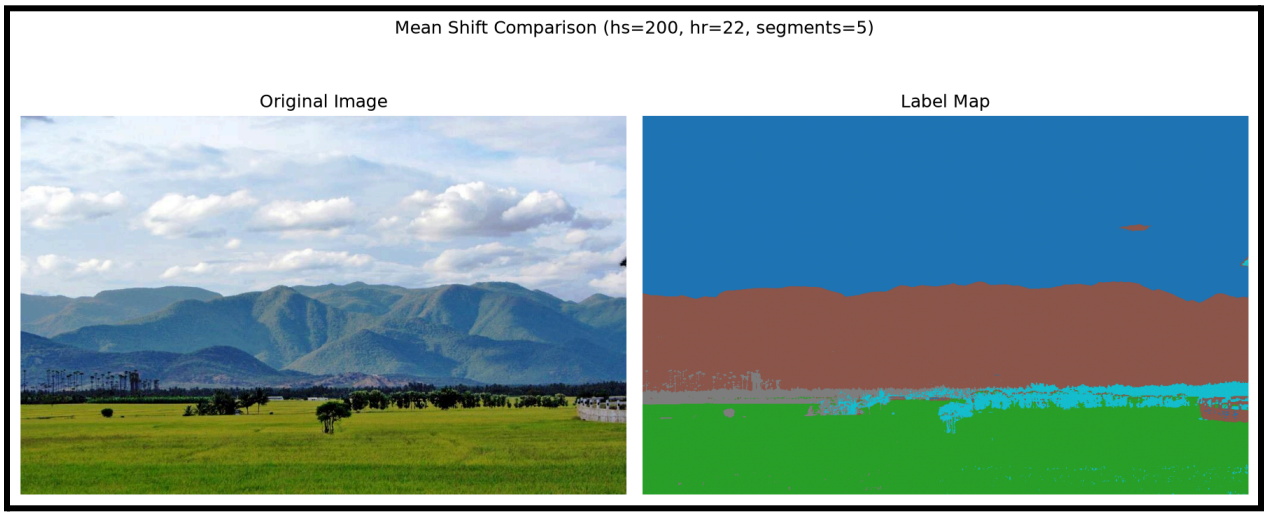
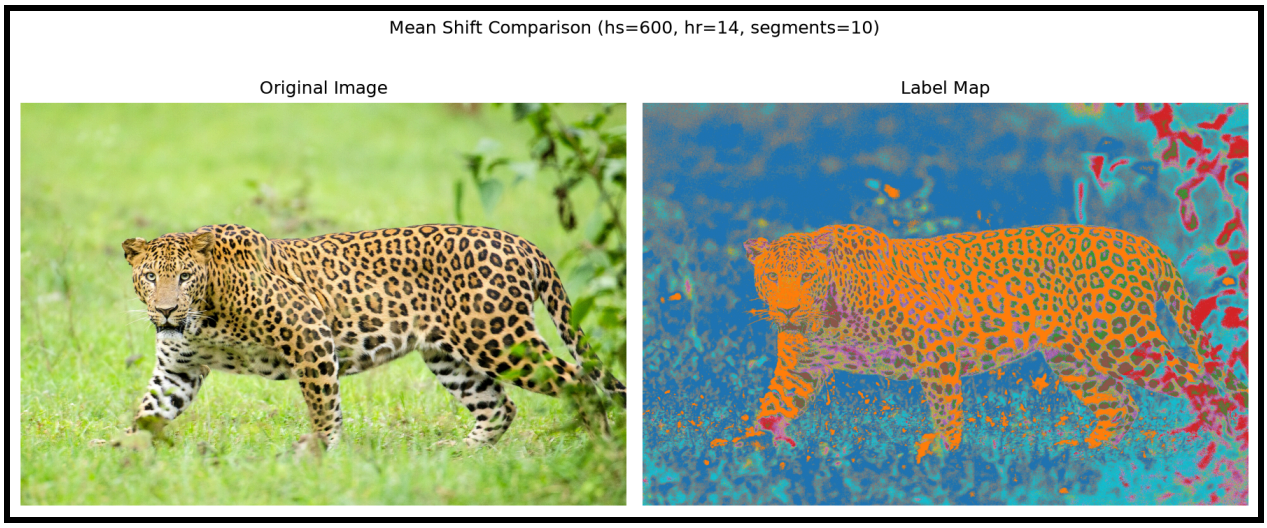
1. Image Segmentation using Mean Shift



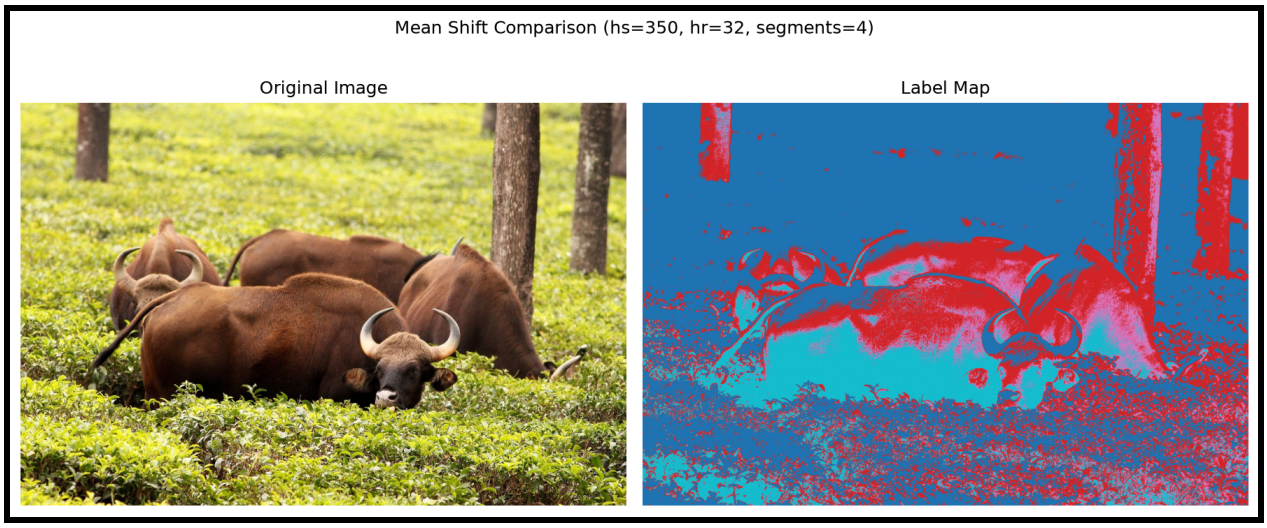
Explanation:

1. ghats.png: 5 segments (hs = 200 & hr = 22)

When mean-shift segmentation is applied to ghats.png, it groups the image into five major regions, including the sky, clouds, mountains, and foreground vegetation. The large homogeneous regions are segmented effectively, but fine details such as individual clouds and variations within the mountain ranges are lost.

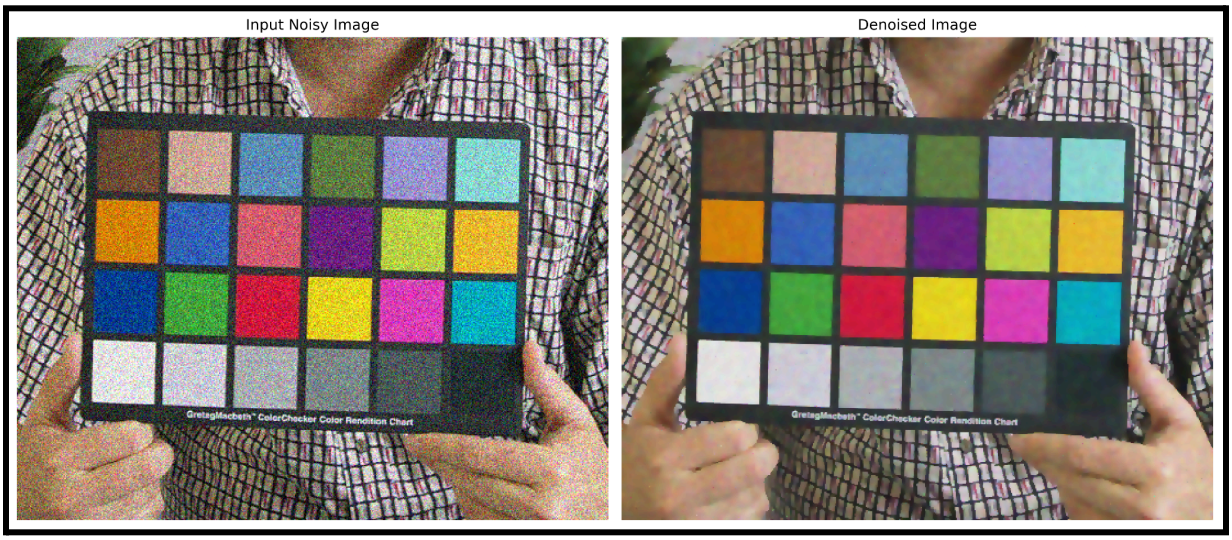


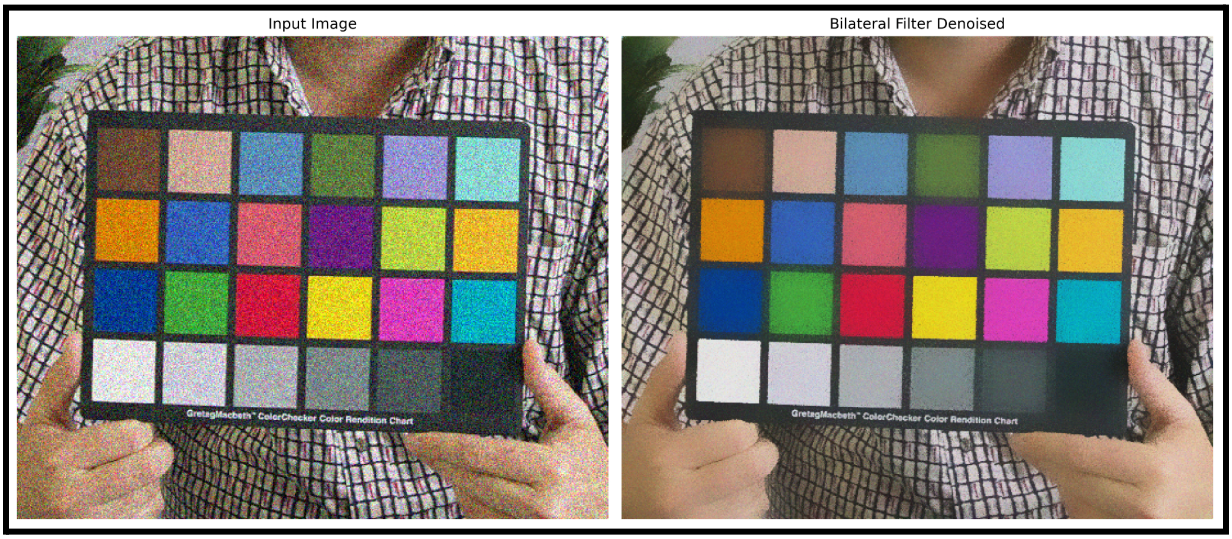
2. leopard.png: 10 segments (hs = 600 & hr = 14)
When the algorithm is applied to leopard.png, it separates the leopard from the grassy background recognizing the characteristic spots and body parts of the animal. While segmentation does a good job in preserving the relevant details of the image through the use of small color bandwidth, the numerous small color changes cause fragmentation as present in the boundary picture.



3. bison.png: 4 segments (hs = 350 & hr = 32)
The algorithm produces four broad colour-based regions, separating the animals, vegetation, and background into dominant clusters. However, the relatively large colour bandwidth merges visually different regions. The boundary image reveals that many fine details within the animals and vegetation are lost, and irregular black contours appear throughout the textured foreground.

2. Anisotropic Diffusion, Bilateral Filtering





Explanation:

1. colorCheckerNoisy.png

Perona Malik Diffusion (K = 20 & lambda = 0.1)

In this algorithm, the parameter (K=20) determines how readily smoothing occurs across intensity changes (gradients, especially when edges appear in the image) and the parameter (lambda=0.1) controls the diffusion step size at each iteration.
In the denoised image, the random noise is reduced across the colour patches, hands, and shirt and at the same time, it attempts to preserve important edges while it smoothes the relatively uniform areas.

Bilateral Filtering (σ_s = 20 & σ_r = 60)

In this algorithm, the spatial parameter (σ_s = 20) controls the influence of nearby pixels, while the range parameter (σ_r = 60) determines how strongly pixels with different intensities influence one another.

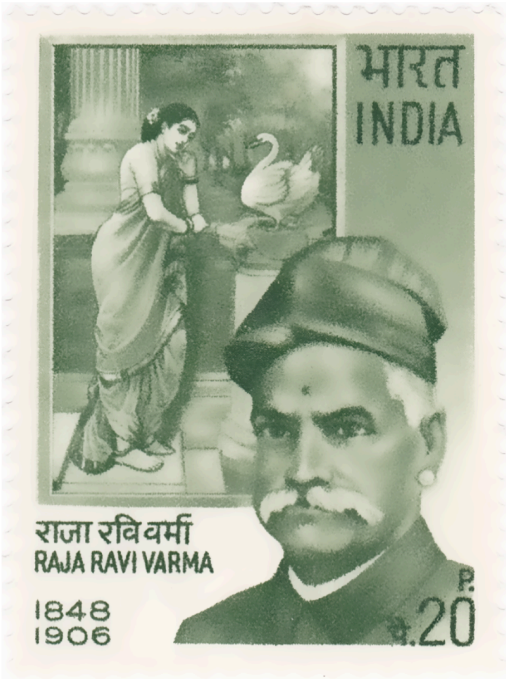
In the denoised image, the random colour noise is substantially reduced, particularly within the colour patches of the ColorChecker chart and the skin regions. However, some fine/grainy texture and small details are still present in the output around the edges.

Input Noisy Image

Denoised Image

Input Image

Bilateral Filter Denoised



2. rajaRaviVarma.png

Perona Malik Diffusion (K = 30 & lambda = 0.05)

As seen in the denoised image, the grainy noise is reduced across the stamp, especially in the background area and in the green areas, which are fairly uniform in their distribution. The text, portraits, borders of the illustrated figures, etc. are still fairly distinguishable because the filter limits smoothing across strong intensity differences.

Bilateral Filtering (σ_s = 10 & σ_r = 45)

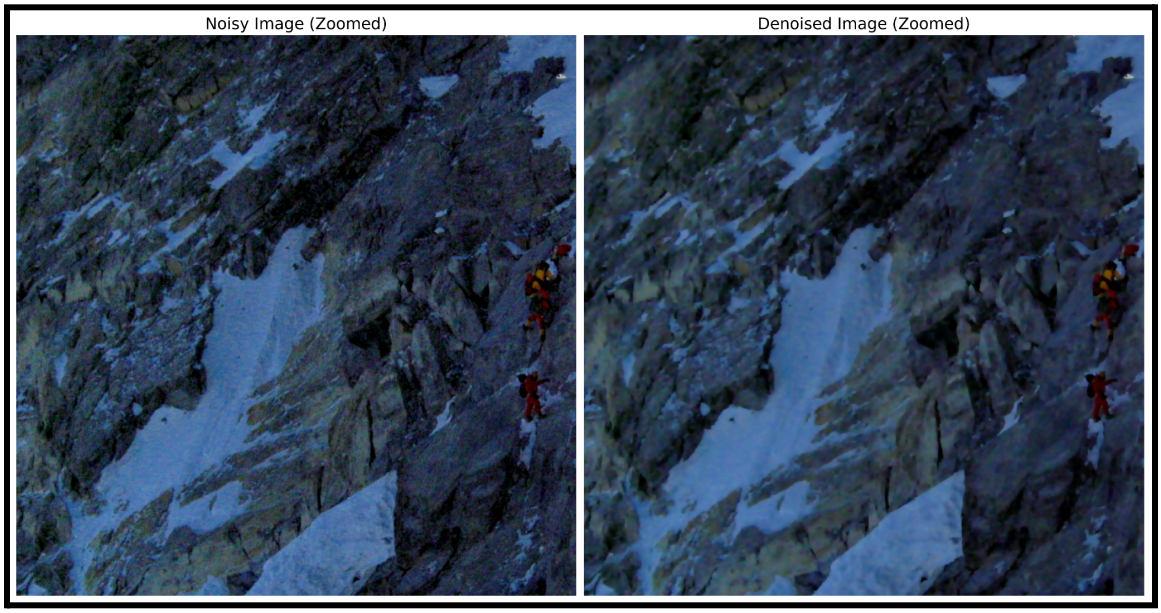
In the denoised image, noise is reduced across the stamp's background, portraits, and illustrated regions. The method attempts to smooth relatively uniform areas while retaining important structures, such as the lettering, portrait contours, and rectangular frame. Nevertheless, some fine image details are softened.

3. sagarmaathaa.png

Original Image

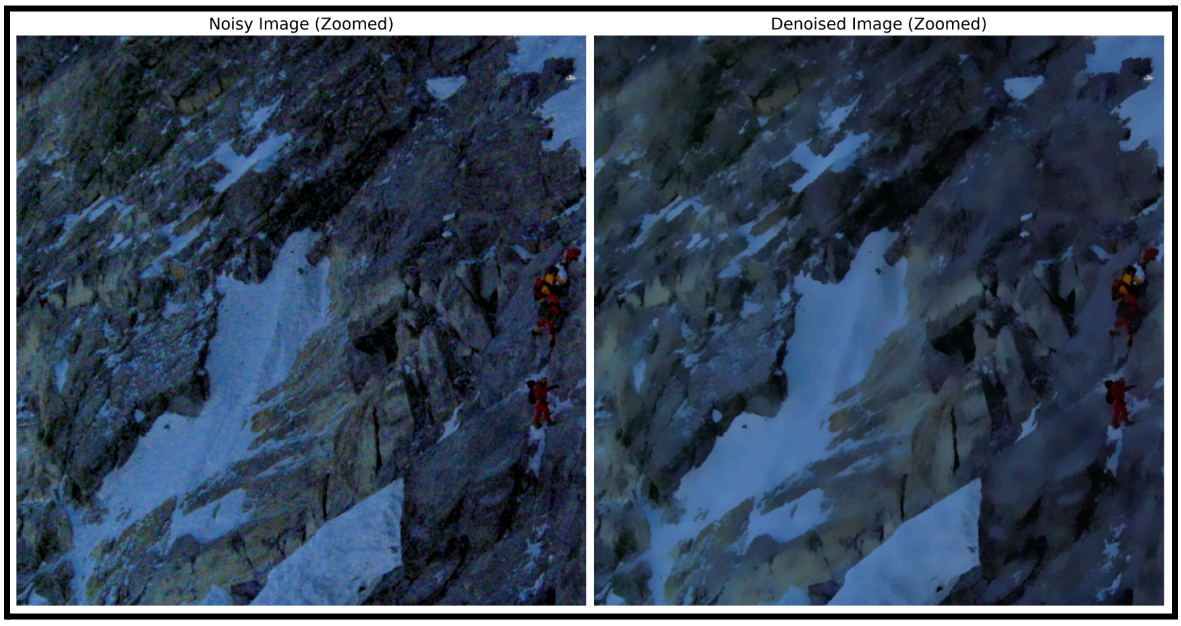
Perona–Malik anisotropic diffusion (top) & Bilateral Filtering (bottom)

Perona Malik Diffusion (K = 15 & lambda = 0.05)



In the zoomed mountain image, the P-M algorithm smooths the snowy areas and parts of the rocky surface while preserving many rock boundaries and the climbers' outlines. However, some fine textures remain, and the image retains a slightly grainy appearance.

Bilateral Filtering (σ_s = 10 & σ_r = 30)



The result in Bilateral filtering appears smoother, particularly across the snow and darker rocky regions, while retaining the major rock formations and climbers. However, some fine rock textures are softened, producing a slightly blurred and cartoonish appearance.

4. racing.png

Original (top) & Perona–Malik anisotropic diffusion (bottom)

Bilateral Filtering

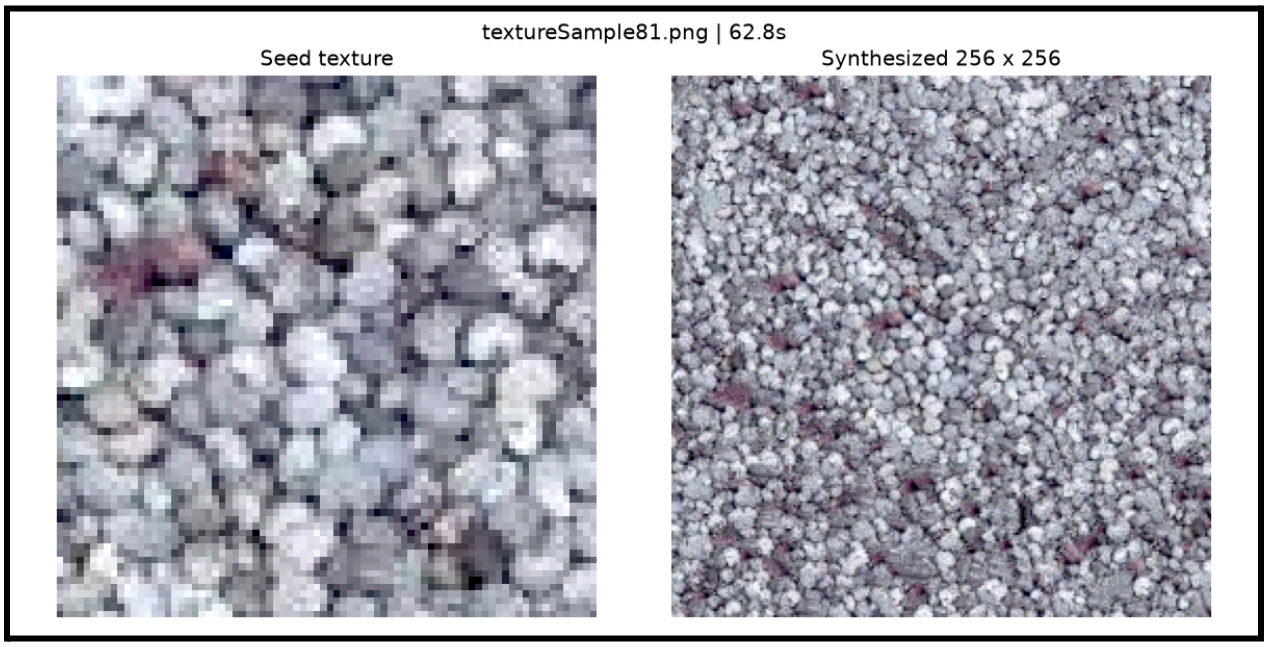
Perona Malik Diffusion (K = 20 & lambda = 0.15)

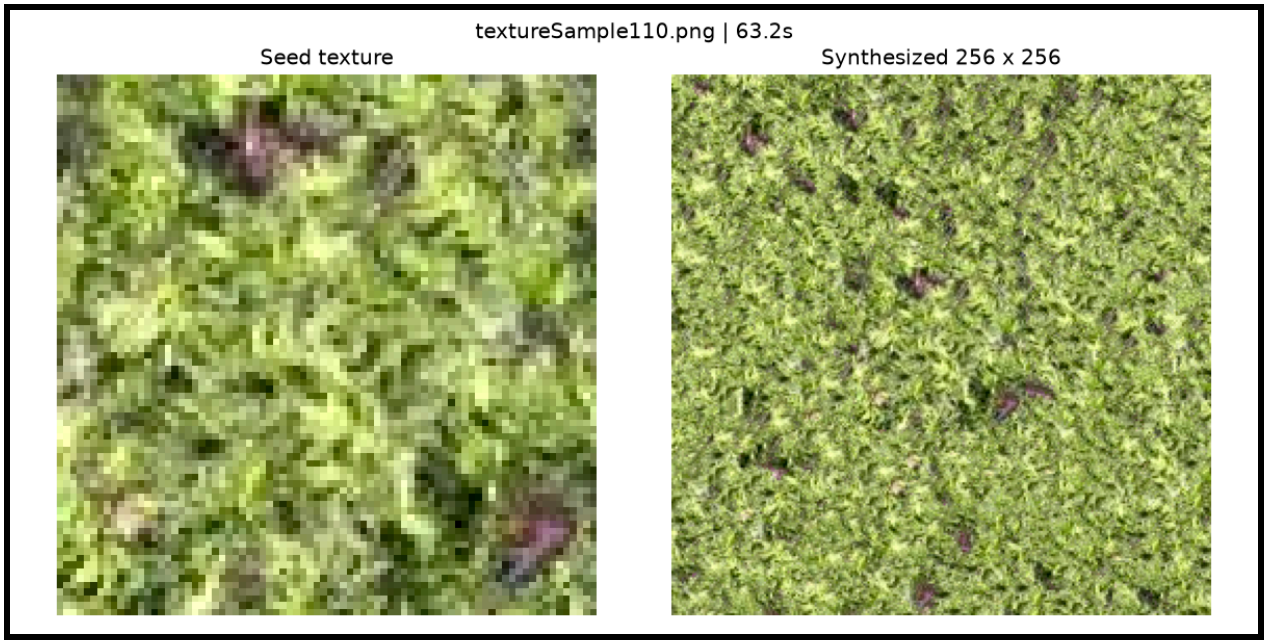
In the zoomed racing image, this method reduces the prominent grainy noise across the dark background, race car, and advertising board. The car's major edges, lettering, and colour transitions remain visible, although some fine details and background textures become smoother.

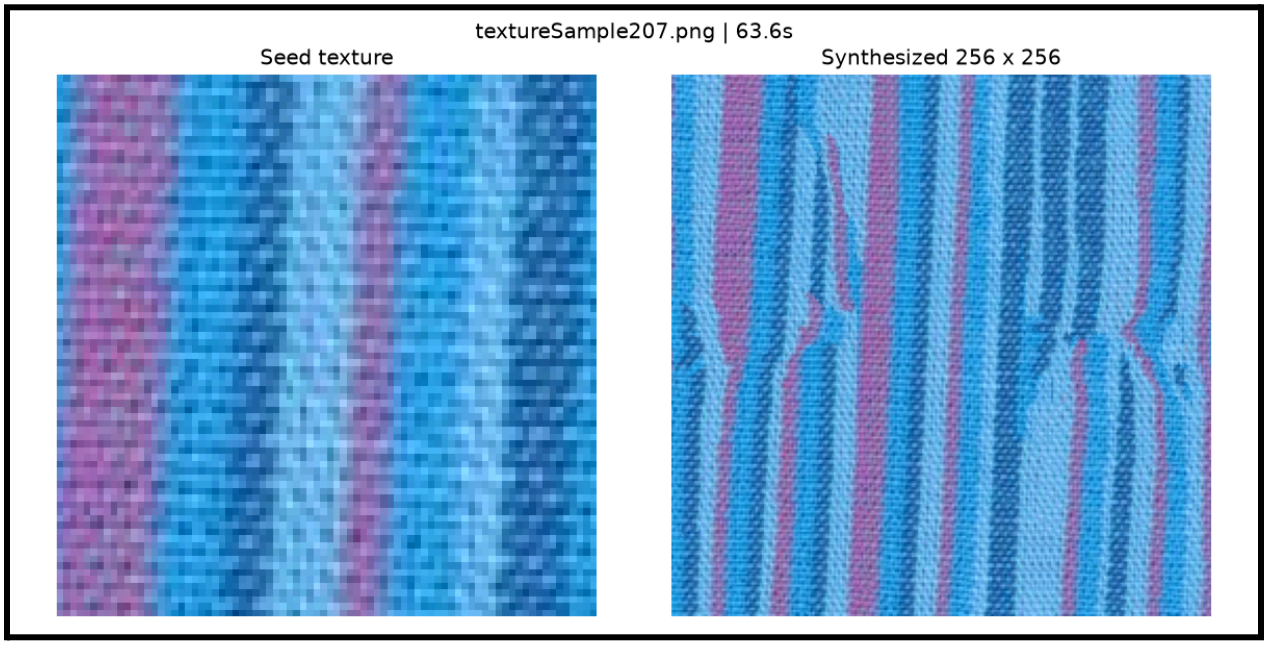
Bilateral Filtering (σ_s = 12 & σ_r = 45)

The denoised image exhibits reduced graininess across the car, background, and advertising board. The car's overall shape and lettering remain distinguishable, but some fine textures and edges appear softer due to the averaging process.

3. Texture Synthesis, Inpainting







Explanation:

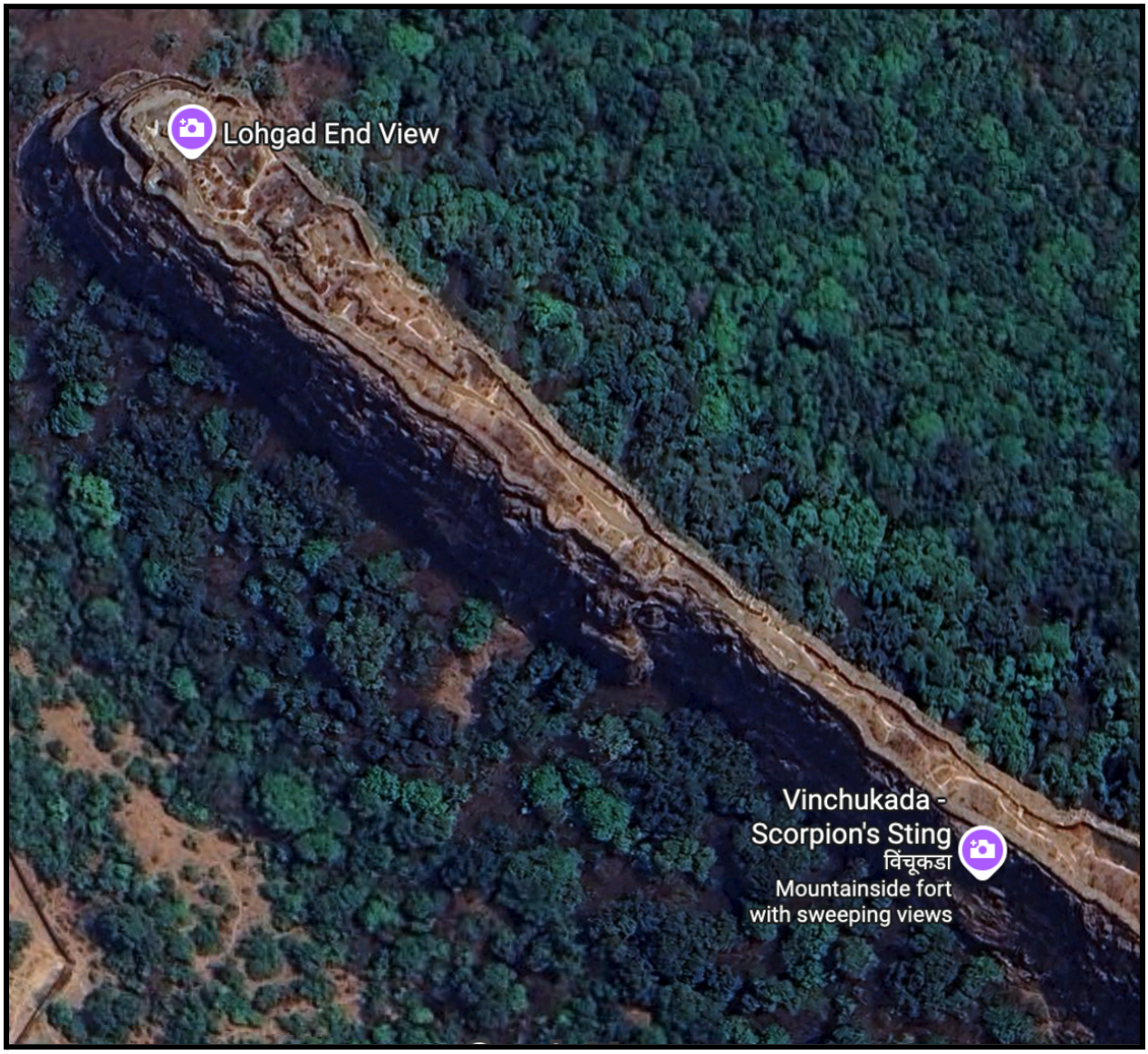
When texture synthesis algorithm is applied to these images:

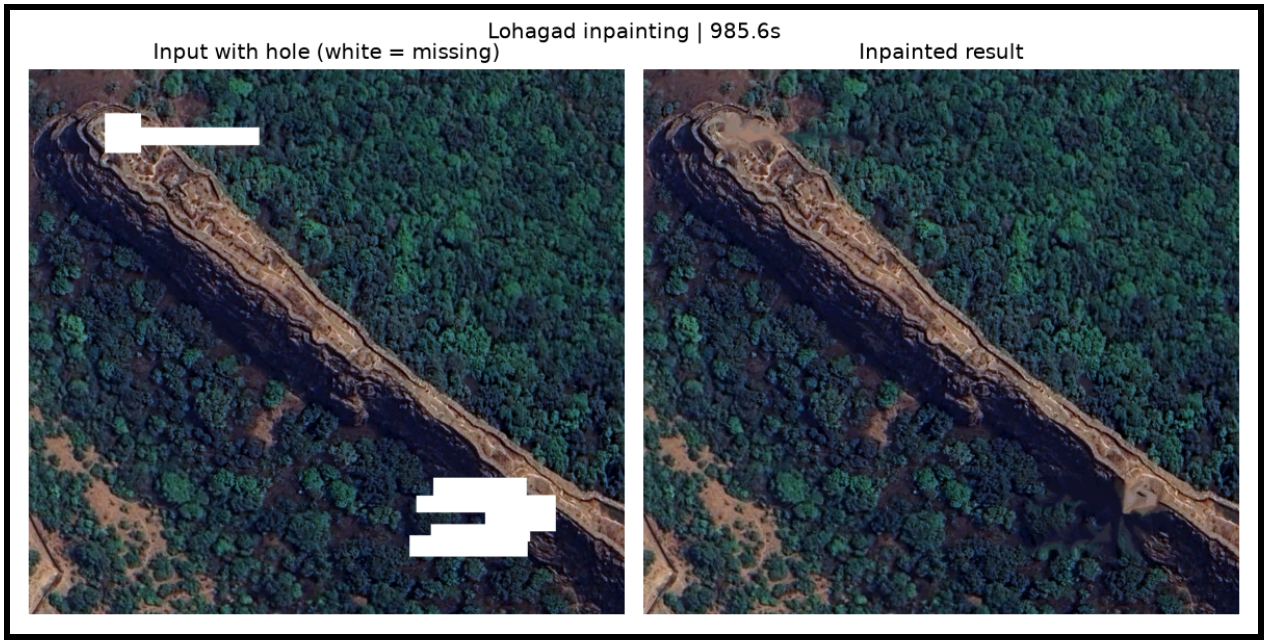
1. textureSample81.png

The synthesized image captures the density of small and round pebbles while keeping intact their greyness, whiteness, and darker color tones. The output preserves the irregularity and granular quality of the original texture creating a visual continuity in the pebble surface.

2. textureSample110.png
The synthesized image perfectly reproduces the dense texture of green foliage, keeping intact its color variations and irregularity of leaf-like structures. The texture looks consistent throughout the output with some darker and purple spots.

3. textureSample207.png
The synthesized image keeps the predominant blue, cyan and pink stripes with the fine woven texture of the fabric. Even though some stripes show some irregularities and discontinuity, still the output stripes keep their original appearance.

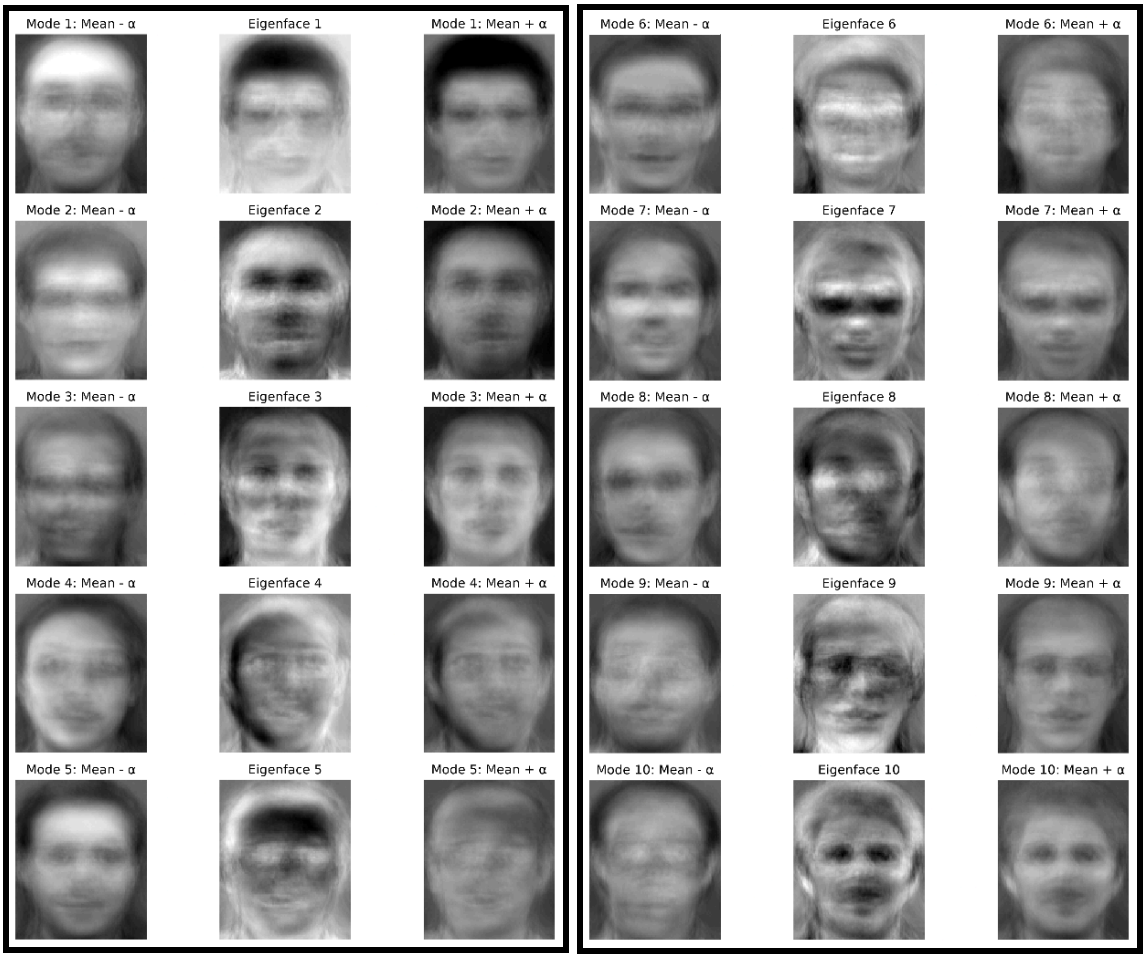




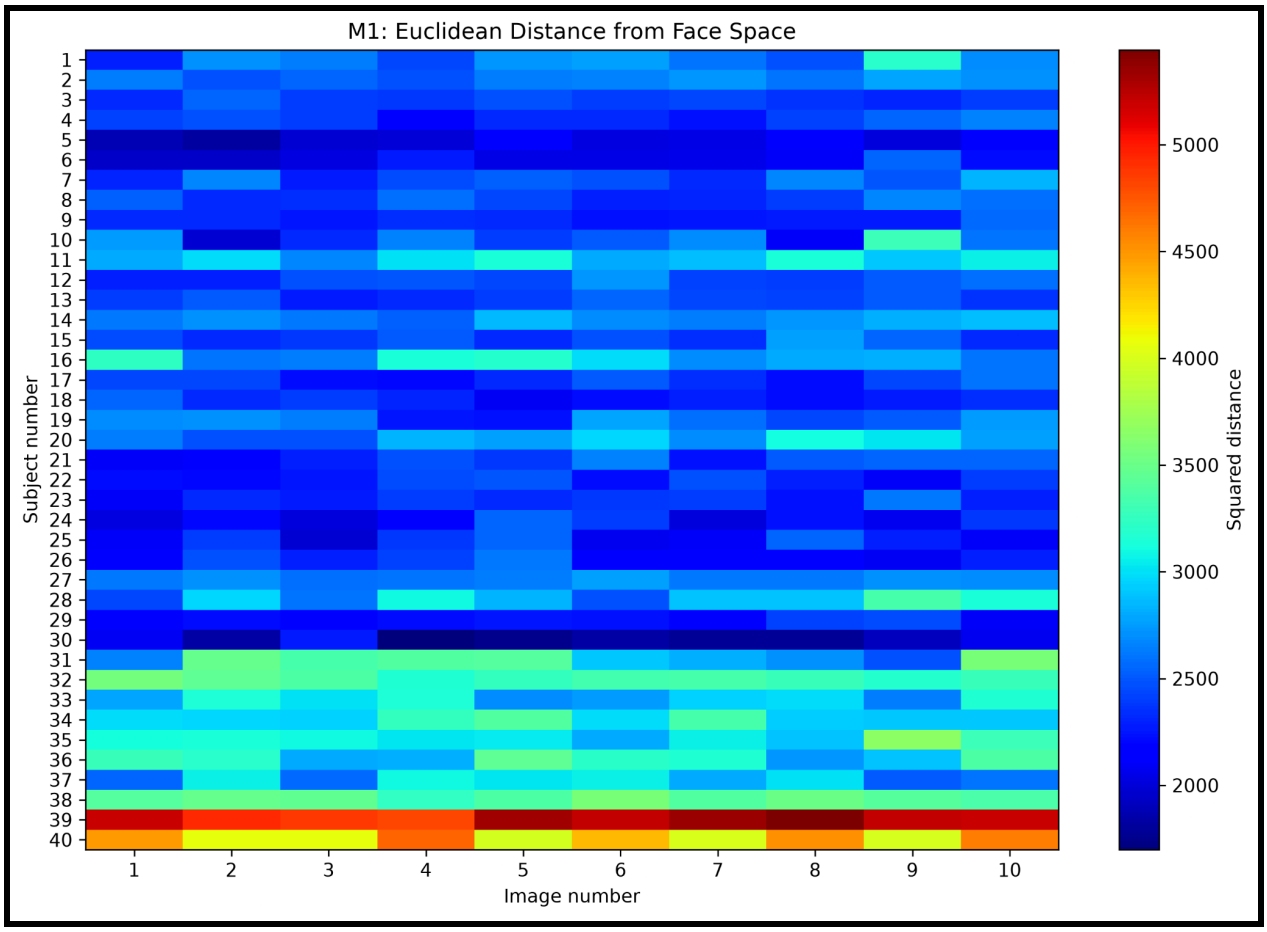
The inpainting algorithm was used to remove Google Maps annotations, including location labels and pins, from the Lohagad aerial image. The masked regions were reconstructed using the surrounding terrain and vegetation as reference. The upper region blends reasonably well with the rocky fort structure, while the lower region exhibits some distortion near the ridge boundary and local terrain details.

4. PCA for face recognition

The average face is derived by finding the pixel wise mean of all the faces from the database after alignment. It signifies the shared facial configuration among all the faces in the database, the position of the eyes, nose, and mouth and also the face structure.

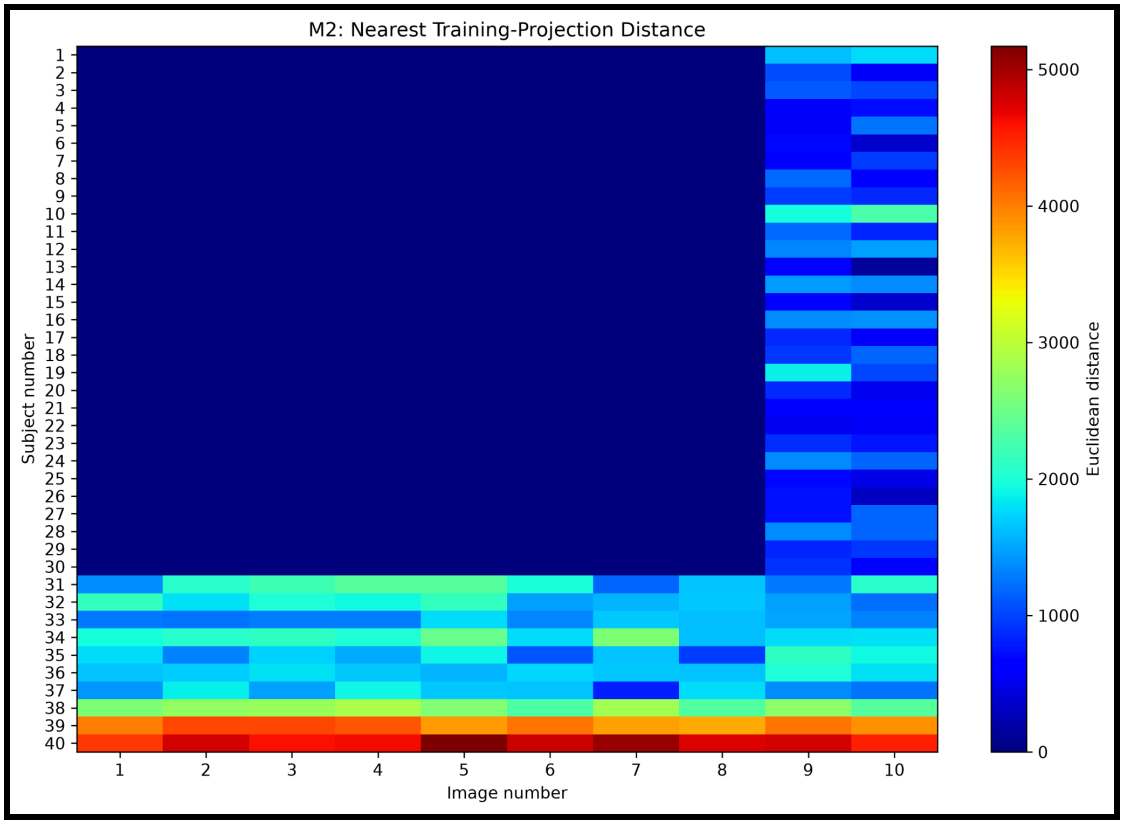


The figure provides the first ten eigenfaces which were generated from PCA. In essence, each one of the eigenfaces is a direction in which the variations exist in the facial data where earlier components capture greater variances compared to later components. The middle column shows the eigenfaces, while the left and right columns show reconstructed faces obtained by subtracting and adding a scaled eigenface to the mean face, respectively (Mean - λvi and Mean - λvi) which highlights the faces of two different people.



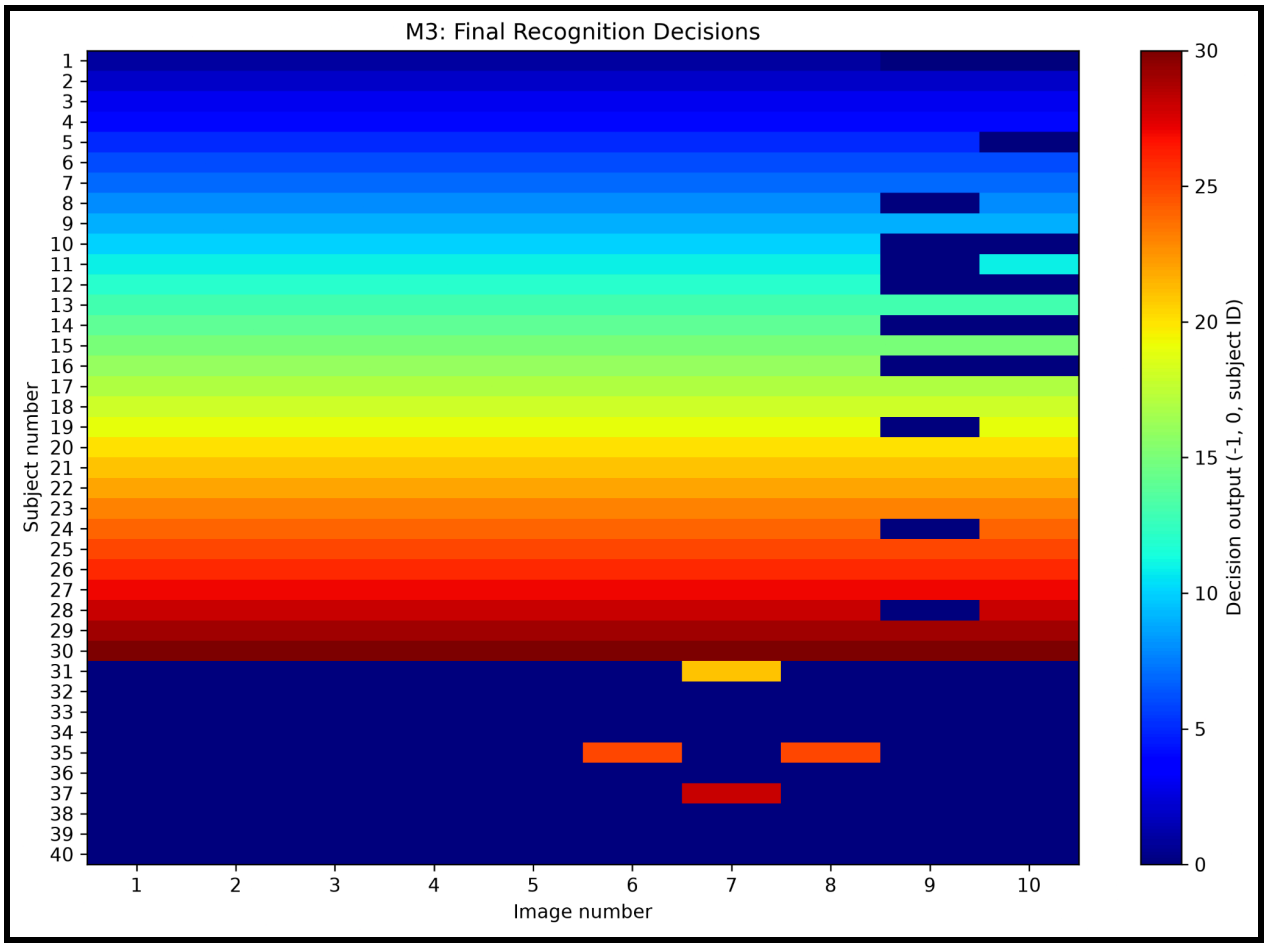
The M1 heatmap represents the distance of each image from the PCA face space. This measure indicates how well an image can be represented using the retained eigenfaces.

The images of most of the people in categories 1-30 have relatively small M1 values, indicating their reasonable representation within the face space that was learned during the training process. In contrast, the images of some people in categories 31-40 have noticeably larger M1 values; specifically, the images of subjects 39 and 40 have relatively high M1 values.



The M2 heatmap shows the minimum Euclidean distance between each image's PCA projection and the projections of the training images.

The distance between the projections of the training images is zero because the training images are part of the gallery against which their projections are being matched. For the rest of the images belonging to known individuals, the distance varies significantly, which suggests that the projections are not always close to the training images. The images of new individuals tend to have larger distances, although there are variations between subjects and individual images.



The M3 heat map indicates the result of classification output for each picture.

From the heatmap, it can be seen that most of the images of unknown individuals are classified as output 0, which means that the classifier was able to reject unknown faces. But there are images from known individuals that were classified as output 0 as well, which means that the classifier was unable to detect the known individual. There are some images from unknown faces that have known face labels.

```
Loading PCA model from pca_model_artifacts.npz...
Loading dataset images...
Loaded images: 400
Training images: 240
Test images: 160
Computing M1 and M2 for all images...

Manually supplied parameters
K: 10
FACE_THRESHOLD: 5500
MATCH_THRESHOLD: 1200.0
Test decision accuracy: 87.50%

Test-set decision counts:
Total: 160
Correct: 140
Incorrect: 20
Output -1: 0
Output 0: 112
Output 1..30: 48

Separate Test Accuracies:
Known-person recognition: 73.33%
Unknown-person rejection: 96.00%
```

The accuracy of the decision process for 87.50% implies that out of 160 testing images, 140 have been assigned the correct three-way decision label. The accuracy of known-person recognition is 73.33%, implying that the classifier could recognize approximately three-fourths of the test images which belong to the subjects whose faces were seen in the training set. The accuracy of unknown-person rejection is 96.00%, meaning that most of the images which belong to new subjects have been properly rejected.